

GRADE 1: Week 1: Phase 1: Warm-up & Class Management (5 Minutes)

Good morning, **children!** Welcome to the **computer lab!**

Please stand next to your **chairs**. Hands up high! Hands down low! Now, **quietly** sit down. Perfect!

Look at me, **everyone**. Today, we have **rules**:

1. Eyes on me.
2. Hands off the **keyboard** for now.
3. **Listen** carefully.

If you are **ready**, give me a big smile! Great! Let's start our **lesson**.

Phase 2: Turning On and Off the Computer (15 Minutes)

Teacher Demonstration (5 mins):

Look at the **screen** in front of you. Is it dark? Yes! That means the computer is **off**.

To turn it **on**, we need to find the **power button**. Look at the **monitor**. Do you see a small round button with a line? That is the **power button**!

Watch me. I **press** it once... click! Now, the **screen** lights up!

Now, look at how we **turn off** the computer. We do **not** just press the power button again. We use the **mouse**! We click the **Start menu** on the bottom left corner, click **Power**, and click **Shut down**. The screen goes **dark**!

Student Practice (10 mins):

Now it's your turn! Put your index finger on the **power button**. Don't press it yet!

On the count of three, **press** it gently. One, two, three... **press**!

Awesome! Your **screens** are turning **on**!

Now, let's practice turning them **off** correctly. Move your **cursor** to the bottom left. Click **Start**, click **Power**, and click **Shut down**.

Great job! Now **press** the **power button** one more time to turn them back **on** for our game!

Phase 3: Learning Mouse Skills & Playing Whack-a-Mole (30 Minutes)

Mouse Technique (5 mins):

Now, look at your **mouse**. It looks like a little **animal**!

Hold the **mouse** gently with your right hand. Put your **index finger** on the **left button**. This is the **left-click**.

Try to **click** it once. *Click!* Try to **double-click** fast. *Click-click!*

When you move your hand, the small arrow on the screen moves. That arrow is called the **cursor**. Move your **cursor** up, down, left, and right!

Game Demonstration (5 mins):

Today, we are playing a game on **plays.org/whack-a-mole!**

Look at my screen. Do you see the **moles** popping up out of the holes?

Your job is to move your **cursor** over the **mole** and **left-click** to hit it! Fast **clicking** earns more **points**. Watch me hit one... *Click!* Got it!

Guided Play (20 mins):

Open your **browser** and go to the **website**.

Ready? Hands on your **mouse**. Start the **game!**

Move your **cursor** quickly! **Click** the **mole!** Good job, Timmy! Nice **double-click**, Sarah!

If a **mole** disappears, find another one fast. Keep **clicking!** You are doing an **amazing** job!

Three, two, one... time is up! Look at your final **score!** Who got more than ten **points?**
Raise your hand! Fantastic work!

Phase 4: Wrap-up & Classroom Cleanup (5 Minutes)

Okay, **children**, **stop** clicking and **freeze!** Hands on your laps.

You did a **wonderful** job using the **mouse** today!

Now, it is time to clean up before we leave.

1. First, let's **shut down** our computers. Move your **cursor** to **Start**, click **Power**, and click **Shut down**. Wait until the **screen** turns black.
2. Second, place your **mouse** neatly next to the **keyboard**.
3. Third, push your **chairs** under the desk **quietly**.

Check your desk. Is everything **clean** and **tidy?** Excellent!

Line up at the door **silently**. Goodbye, **class!** See you next week!

Guide to Key Terms & Pronunciation for Teachers

Below are the **keywords** and **difficult-to-pronounce words** from the lesson script, along with their simple phonetics and Vietnamese meanings:

English Term	Pronunciation (IPA)	Vietnamese Meaning
Keyboard	/ˈkiː.bɔːrd/	Bàn phím
Monitor	/ˈmɑː.nə.t̬ə/	Màn hình
Power button	/ˈpaʊ.ə ˌbʌt.ən/	Nút nguồn
Cursor	/ˈkɜː.sə/	Con trỏ chuột
Double-click	/ˈdʌb.əl.klɪk/	Nhấp cú pháp đôi (click 2 lần)
Browser	/ˈbraʊ.zə/	Trình duyệt web
Mole	/moʊl/	Con chuột chũi
Quietly	/ˈkwaɪ.ət.li/	Một cách giữ trật tự
Shut down	/ʃʌt daʊn/	Tắt máy tính
Disappears	/ˌdɪs.əˈpiːrɪz/	Biến mất
Freeze	/friːz/	Bất động / Giữ nguyên tư thế
Tidy	/ˈtaɪ.di/	Gọn gàng

GRADE 2: Week 1

PHASE 1: CLASSROOM MANAGEMENT & WARM-UP (5 MINUTES)

Good morning, **everyone**! Welcome to today's **computer class**!

Standing up, please! Hands up to the **ceiling**! Hands down to your **toes**! Now, sit down **quietly** and put your hands on your laps. Excellent!

Look at me, **class**. Today, we have a very exciting **mission**, but first, let's **review** our room rules:

1. Eyes on the **teacher**.
2. Keep your hands off the **keyboard** and **mouse** until I say go.
3. Keep the room **peaceful**.

Are we ready? Show me your big **thumbs-up**! Perfect, let's start!

PHASE 2: PAGE 19 & DOJO ISLAND INTRO (15 MINUTES)

Open your **textbook** to **page nineteen** (19). That is a **one** and a **nine**.

Look at the **picture** on **page nineteen**. What game is that? Yes, it looks like **Minecraft**! But today, we are **not** building with **Minecraft blocks**. We are going to build **together** on **Dojo Island**!

Listen **carefully** to the **instructions** on how to get there:

1. **First**, find the **Chrome browser** icon on your **desktop**. It looks like a colorful circle—red, yellow, green, and blue. **Double-click** it to open!
2. **Second**, look at the **address bar** at the top. Type **d-o-j-o-dot-m-e (dojo.me)** and press **Enter** on your **keyboard**.
3. **Third**, you will see a list of names. Scroll down and **locate** your own **fullname**. Click on your name to log in!

Now, it's your turn! Open **Chrome**, type **dojo.me**, and find your name. Go!

(Teacher walks around silently monitoring)

Great job! Now that you are on **Dojo Island**, how do we move our **characters**? Look at your **keyboard**. Find the **arrow keys**—**up**, **down**, **left**, and **right**. Press the **up arrow** to walk forward, and the **left arrow** to turn left. Try moving your **avatar** right now!

PHASE 3: BUILDING ON DOJO ISLAND & RULES (30 MINUTES)

Now comes the most **exciting** part! We are going to **collaborate** and **build** houses together on the island!

Before we start, listen to this very **important rule**:

How many **teachers** do you have in this room? Only **one** or **two**, right? Yes, **only** one or two! I cannot answer thirty **questions** at the same time.

So, if you have a **question** or if you get **stuck**:

- **Option one:** Ask your **neighbor** sitting next to you quietly.
- **Option two:** Raise your hand **patiently** and wait for me to come.

Do **not** shout my name! Do **not** make loud **noises**!

Look at this **lock icon** on my screen. If the class gets too **noisy** and **disruptive**, **Dojo Island** will be **locked** immediately! If it gets **locked**, nobody can play or **build** anymore, and we will switch to doing written **exercises** in our notebook.

Keep your voices low, work **peacefully**, and help each other.

You have 20 minutes to build your **structures**. Use your **arrow keys** to explore, gather materials, and build! Start **building**!

Students play for 20 minutes.

Teacher walks around, reinforcing quiet voice and peer help.

Three, two, one... time is up! Take your hands off the **mouse** and **keyboard**! Look at all the **amazing** houses you built **together**! Give yourselves a **pat on the back**!

PHASE 4: CLEANUP & DISMISSAL (5 MINUTES)

Okay, **children**, **freeze**! It is time to clean up before we leave the **lab**.

Follow my **steps**:

1. Click the **close button**—the red **X** at the top right corner—to close the **Chrome browser**.
2. Click **Start**, click **Power**, and click **Shut down** to turn off your **computers**.
3. Place your **mouse** neatly next to the **keyboard**. Put your **textbook** back in your **backpack**.
4. Push your **chairs** in **silently**.

Check your desk area. Is there any **trash**? Is everything **organized** and **tidy**?

Line up at the door **quietly** in a single file. You all did a **fantastic** job today working together on **Dojo Island**! See you next time! Goodbye, **class**!

TEACHER'S GLOSSARY

Từ tiếng Anh	Phiên âm quốc tế (IPA)	Ý nghĩa tiếng Việt	Ghi chú phát âm
Ceiling	/ˈsiː.lɪŋ/	Trần nhà	Chú ý âm /s/ đầu và âm /ɪŋ/ cuối.
Quietly	/ˈkwaɪ.ət.li/	Một cách giữ trật tự	Phát âm 3 âm tiết, tránh đọc thành "qui-et-ly".
Peaceful	/ˈpiːs.fəl/	Hòa bình / Yên tĩnh	Âm /iː/ kéo dài, đuôi /-fəl/ nhẹ.
Page nineteen	/peɪdʒ ˌnaɪnˈtiːn/	Trang 19	Nhấn trọng âm vào "TEEN" (19), tránh nhầm với "FORTY" (40).
Instructions	/ɪnˈstrʌk.ʃənz/	Hướng dẫn	Trọng âm vào âm tiết thứ 2 (STRUK).
Browser	/ˈbraʊ.zə/	Trình duyệt web	Âm /aʊ/ giống như "bao" trong tiếng Việt.
Locate	/ləʊˈkeɪt/	Tìm thấy / Xác định	Trọng âm rơi vào âm tiết thứ 2.
Fullname	/fʊl neɪm/	Họ và tên	Đọc rõ âm /l/ ở cuối từ "full".
Arrow keys	/ˈær.əʊ kiːz/	Phím mũi tên	Chú ý âm /r/ uốn lười.
Character	/ˈkær.ək.tə/	Nhân vật	Bắt đầu bằng âm /k/ (không đọc là /tʃ/).
Avatar	/ˈæv.ə.tɑːr/	Hình đại diện / Nhân vật game	Trọng âm rơi vào âm tiết đầu tiên.

Từ tiếng Anh	Phiên âm quốc tế (IPA)	Ý nghĩa tiếng Việt	Ghi chú phát âm
Collaborate	/kə'ləb.ə.reɪt/	Hợp tác / Làm việc nhóm	Trọng âm rơi vào âm tiết thứ 2 (LAB).
Disruptive	/dɪs'rʌp.tɪv/	Bất trật tự / Gây rối	Trọng âm rơi vào âm tiết thứ 2 (RUP).
Immediately	/ɪ'miː.di.ət.li/	Ngay lập tức	Chú ý phát âm đủ 5 âm tiết.
Exercises	/'ek.sə.saɪ.zɪz/	Bài tập	Đuôi /-zɪz/ phát âm rõ ràng.
Organized	/'ɔːr.gə.naɪzɪd/	Ngăn nắp	Trọng âm rơi vào âm tiết đầu tiên.

GRADE 3: Week 1:

PHASE 1: CLASSROOM MANAGEMENT & WARM-UP (5 MINUTES)

Good morning, **third-graders**! Welcome back to **computer science class**!

Standing up straight, please! Hands up to the **sky**! Hands out to the **sides**! Now, sit down **quietly** and put your hands on your desks. Excellent!

Let's quickly review our three **classroom rules** for today:

1. Eyes on the **instructor**.
2. Keep your hands off the **keyboard** and **mouse** when I am speaking.
3. Be **respectful** and keep the room **peaceful**.

Are we **ready** to start? Show me a big **thumbs-up**! Perfect, let's begin!

PHASE 2: TEXTBOOK TASK & MAP CODING (20 MINUTES)

Open your **textbooks** to **page twenty-four** (24). That is a **two** and a **four**.

Look at the **page**. You will see a total of **three maps**: Map 1, Map 2, and Map 3!

Listen **carefully** to your **mission**:

1. You need to **run code** by following the **directional arrows**—up, down, left, and right.
2. Follow the **arrows** on each map to reach the **destination** and answer the **questions** in your **textbook**.

Let's do Map 1 **together** as an **example**. Look at the first **arrow**. It points **right**, so we move one **step** to the **right**. The second **arrow** points **up**, so we move one **step upward**.

Now, complete Map 2 and Map 3 on your own! Use your **pencil** to trace the **path** and write down the **answers**. You have 10 minutes. Begin!

(Teacher walks around silently, checking work and assisting students)

Three, two, one... time is up! Put your **pencils** down. Excellent job **decoding** those paths, **everyone**!

PHASE 3: ONLINE CODING ON CODE.ORG (30 MINUTES)

Now, we are going to transition to our online **coding workspace**!

Follow my **instructions** step-by-step on your **computers**:

1. Open your **Chrome browser**.
2. In the **address bar**, type **c-o-d-e-dot-o-r-g** (**code.org**) and press **Enter** on your **keyboard**.
3. Look at the top right **corner**. Click the **Sign in** button!

4. You will see a box asking for a **section code**. Type our six-letter **classroom code**: **A-B-C-D-E-F**, then click **Go**!
5. Now, look at the screen. You will see a list of **avatars** and **names**. **Locate** your **fullname** and click on your **character**.
6. Type your secret **picture password** or **text password** to log in!

Once you are in, **code.org** will show you step-by-step **puzzles**. Read the **instructions** at the top of the **screen** and drag the **blocks** to make your **character** move!

If you get **stuck**, don't panic! **Read** the instructions again or ask your **neighbor quietly**. If you still need help, raise your hand **patiently**.

You have 20 minutes to solve as many **coding puzzles** as you can. Go, **programmers**!

(Students complete puzzles on code.org while teacher monitors)

Great effort, **class**! Look at all those green **completed levels**! Give yourselves a **round of applause**!

PHASE 4: CLEANUP & DISMISSAL (5 MINUTES)

Okay, **children**, **freeze**! Stop **clicking** and put your hands on your laps.

It is time to clean up our **workstation** before leaving:

1. Click the red **X** at the top right **corner** to close the **browser**.
2. Click **Start**, click **Power**, and click **Shut down** to turn off your **computers**.
3. Close your **textbooks** and put your **pencils** away inside your **backpacks**.
4. Push your **chairs** in **silently** under the desk.

Is your area clean and **tidy**?

Line up at the door **silently**. You all worked like real **computer programmers** today! See you next week! Goodbye, **class**!

TEACHER'S GLOSSARY

Instructor	/ɪnˈstrʌk.tə-/	Người hướng dẫn / Giáo viên	Trọng âm rơi vào âm tiết thứ 2 (STRUK).
Respectful	/rɪˈspekt.fəl/	Tôn trọng	Âm /k/ nhẹ ở giữa, kết thúc bằng /-fəl/.
Peaceful	/ˈpiːs.fəl/	Trật tự / Yên bình	Âm /i:/ kéo dài.

Directional	/də'rek.fən.əl/	Định hướng	Phát âm 4 âm tiết rõ ràng.
Destination	/ˌdes.tə'nei.fən/	Điểm đến	Trọng âm chính rơi vào âm "NAY".
Upward	/ˈʌp.wəd/	Hướng lên trên	Âm /wəd/ phát âm nhẹ, uốn lười.
Decoding	/di:'koud.ɪŋ/	Giải mã / Đọc mã	Trọng âm rơi vào "CODE".
Transition	/træn'ziʃ.ən/	Chuyển sang (hoạt động khác)	Chú ý âm /ziʃ/ ở giữa.
Workspace	/ˈwɜ:k.speɪs/	Không gian làm việc	Phát âm rõ âm /k/ ở giữa và âm /s/ ở cuối.
Instructions	/ɪn'strʌk.fənz/	Hướng dẫn	Trọng âm vào âm tiết thứ 2.
Locate	/lou'keɪt/	Tìm thấy / Xác định vị trí	Trọng âm rơi vào âm tiết thứ 2.
Avatars	/ˈæv.ə.ta:rz/	Hình đại diện Nhân vật	Trọng âm rơi vào âm tiết đầu tiên.
Fullname	/fʊl neɪm/	Họ và tên	Đọc rõ âm /l/ ở cuối từ "full".
Puzzles	/ˈpʌz.əlz/	Các câu đố / Màn chơi	Đuôi /-z/ có âm /z/ rung nhẹ.
Programmers	/ˈprəʊ.græm.əz/	Lập trình viên	Trọng âm rơi vào âm tiết đầu tiên (PRO).
Workstation	/ˈwɜ:kˌsteɪ.fən/	Góc / Màn làm việc	Từ ghép, chú ý âm /k/ giữa hai từ.

GRADE 4 and 5: Week 1:

PHASE 1: WARM-UP, INTRODUCTIONS & CLASSROOM MANAGEMENT (5 MINUTES)

Good morning, **everyone!** Welcome to our **computer science lab!**

Please stand next to your **desks**. Stretch your arms up high to the **ceiling!** Move them down to your **toes!** Now, sit down **quietly** and put your hands on your laps. Excellent!

Since today is our **very first lesson** together, we need to get to know each other! I will walk around, and when I point to you, please say your **fullname** loudly and clearly so I can **input** your information into my **database**.

(Teacher collects and inputs names quickly)

Thank you! Now, let's review our three **classroom rules**:

1. Keep your eyes on the **instructor**.
2. Keep your hands off the **keyboard** and **mouse** when I am speaking.
3. Be **respectful** and work **peacefully**.

Are we ready? Give me a big **thumbs-up!** Perfect, let's begin!

PHASE 2: GOOGLE MEET & SCREEN SHARING SETUP (15 MINUTES)

Today, we are going to use **Google Meet** inside our lab! This allows you to see my **screen share** directly on your own **monitors**.

Follow my steps **carefully**:

1. Open your **Chrome browser**.
2. Type **m-e-e-t-dot-g-o-o-g-l-e-dot-c-o-m** (**meet.google.com**) in the **address bar** and press **Enter**.
3. Click on **Join or start a meeting** and type our meeting **passcode: C-S-L-A-B-1**.
4. Before clicking join, check your display name! You must follow our **naming format**: type your **class**, then a **dash**, and then your **fullname**—for example, **4A - John Smith** or **5B - Sarah Jenkins**. This allows me to **identify** you and read your names easily!
5. Now, click **Join now!**

Once you are inside the **virtual room**, look at your screen. You can now see my **presentation screen share!** When I show **demonstrations**, you can watch them right in front of you.

PHASE 3: TARGET CONCEPTS & CODE.ORG GAMEPLAY (30 MINUTES)

1. Concept Introduction (5 mins):

Look at my **screen share** on **Google Meet**:

- **For Grade 4:** Today, we are learning about **loops** and the **repeat block**! Instead of writing the same code ten times, a **loop** lets us repeat an action automatically! It saves time and makes our **code** clean and **efficient**.
- **For Grade 5:** Today, we are learning about **variables**! A **variable** is like a labeled **container** or **box** that stores information—like a player's **score**, **speed**, or **lives**—that can change while the game is running!

2. Logging into Code.org (5 mins):

Now, open a new **tab** in your **browser**:

1. Go to **c-o-d-e-dot-o-r-g** (**code.org**).
2. Click **Sign in** at the top right **corner**.
3. Type our **section code**: **X-Y-Z-1-2-3** and click **Go**.
4. Select your **fullname** from the list, enter your **secret picture password**, and log in!

3. Self-Paced Practice (20 mins):

You will now solve the **interactive puzzles** on **code.org**.

Remember, everyone learns at their own **pace**! This is not a **race**. If you finish a level quickly, move on to the next **challenge**. If you get **stuck**, don't worry!

- **Option 1: Re-read** the instructions at the top of the screen.
- **Option 2:** Ask your **neighbor** sitting next to you **quietly**.
- **Option 3:** Raise your hand **patiently**, and I will check your screen.

You have 20 minutes. Start **programming**!

(Students work independently while teacher monitors progress on dashboard)

PHASE 4: LESSON SUMMARY & CLEANUP (10 MINUTES)

1. Lesson Summary & Closing (5 mins):

Time is up! Please **freeze** and put your hands on your desks!

Look at my **screen share** on **Google Meet** one last time. I am looking at our **teacher dashboard**, and I am so **impressed**!

- **Grade 4 students:** You did a **phenomenal** job mastering **loops** today! You made your characters **repeat** actions effortlessly!
- **Grade 5 students:** You handled **variables** like true senior **programmers**! You successfully stored and updated data in your games!

Coding is a **journey**, not a sprint. It does not matter which **level** or **stage** you reached today—what matters is that you **persevered**, solved **complex problems**, and learned something new! We will continue right from where you stopped in our next **session**. Give yourselves a big **round of applause**!

2. Classroom Cleanup (5 mins):

Now, let's leave our lab clean and **organized**:

1. Click the red **phone icon** to leave the **Google Meet**.
2. Close your **Chrome browser** by clicking the red **X**.
3. Click **Start**, click **Power**, and click **Shut down** to turn off your **computers**.
4. Place your **mouse** and **keyboard** neatly on the desk.
5. Push your **chairs** in **silently**.

Check your desk for any **trash** or personal items. Line up at the door **quietly**. You were all **extraordinary** today! See you next week! Goodbye, **class**!

TEACHER'S GLOSSARY

Database	/ˈdeɪ.tə.beɪs/	Cơ sở dữ liệu	Trọng âm rơi vào âm tiết đầu tiên (DAY).
Instructor	/ɪnˈstrʌk.tə/	Người hướng dẫn / Giáo viên	Trọng âm rơi vào âm tiết thứ 2 (STRUK).
Respectful	/rɪˈspekt.fəl/	Tôn trọng	âm /k/ nhẹ ở giữa, đuôi /-fəl/ đọc nhẹ.
Peacefully	/ˈpiːs.fəl.i/	Yên bình / Trật tự	Âm /i:/ kéo dài, phát âm 3 âm tiết.
Virtual room	/ˈvɜː.tʃu.əl ru:m/	Phòng học ảo	Chú ý âm /vɜː/ uốn lượn và âm /tʃ/ ở giữa.
Display name	/dɪˈspleɪ neɪm/	Tên hiển thị	Trọng âm rơi vào "PLAY".
Naming format	/ˈneɪ.mɪŋ ˈfɔːr.mæt/	Định dạng đặt tên	Đọc rõ âm /t/ cuối từ "format".

Identify	/aɪ'den.tə.fai/	Nhận diện / Xác định	Trọng âm rơi vào âm tiết thứ 2 (DEN).
Presentation	/,prez.ən'teɪ.ʃən/	Bài thuyết trình / Trình chiếu	Trọng âm chính rơi vào âm tiết thứ 3 (TAY).
Demonstrations	/,dem.ən'streɪ.ʃənz/	Các phần minh họa	Trọng âm chính rơi vào "STRAY".
Loops	/lu:ps/	Vòng lặp (Lớp 4)	Âm /u:/ kéo dài, có âm /s/ ở cuối.
Repeat block	/rɪ'pi:t blɔ:k/	Khối lệnh lặp lại	Chú ý âm /t/ ở "repeat" và /k/ ở "block".
Efficient	/ɪ'fɪʃ.ənt/	Hiệu quả / Tối ưu	Trọng âm rơi vào âm tiết thứ 2 (FISH).
Variables	/'ver.i.ə.bəlz/	Các biến số (Lớp 5)	Trọng âm rơi vào âm tiết đầu tiên (VER).
Container	/kən'teɪ.nə/	Hộp chứa / Thùng chứa	Trọng âm rơi vào âm tiết thứ 2 (TAY).
Interactive	/,ɪn.tə'æk.tɪv/	Tương tác	Trọng âm rơi vào âm tiết thứ 3 (AK).
Pace	/peɪs/	Nhịp độ / Tiến độ	Đọc âm /s/ rõ ràng ở cuối từ.
Challenge	/'tʃæl.ɪndʒ/	Thử thách	Bắt đầu bằng âm /tʃ/, đuôi /-dʒ/ bật nhẹ.
Impressed	/ɪm'prest/	Bị ấn tượng	Đuôi "-ed" phát âm là /t/ (/ɪm'prest/).

Phenomenal	/fə'ni:.mə.nəl/	Xuất sắc / Tuyệt vời	Trọng âm rơi vào âm tiết thứ 2 (NA).
Effortlessly	/'ef.ət.ləs.li/	Một cách dễ dàng	Trọng âm rơi vào âm tiết đầu tiên (EF).
Persevered	/,pɜ:.sə'vɪrd/	Kiên trì / Nỗ lực	Trọng âm rơi vào âm tiết thứ 3 (VEER).
Organized	/'ɔ:r.gə.naɪzd/	Ngăn nắp	Trọng âm rơi vào âm tiết đầu tiên.
Extraordinary	/ɪk'strɔ:r.dən.er.i/	Phi thường / Tuyệt vời	Từ 5 âm tiết, trọng âm rơi vào "STROR".

GRADE 7 and 8: Week 1

- **Grade 7 focus:** Flowcharts from textbook pages 10–16 (transitioning logic into Python programming) before working on codex.io.
- **Grade 8 focus:** Python fundamentals in codex.io, specifically completing the **Hello World** and **Variables** chapters.

PHASE 1: CLASSROOM MANAGEMENT & INTRODUCTIONS (5 MINUTES)

Good morning, **everyone!** Welcome to our **computer science lab!**

Please stand next to your **desks**. Stretch your arms up high to the **ceiling!** Touch your **toes!** Now, take a seat **quietly** and place your hands on your desks. Excellent!

Since today is our **very first lesson** together, I want to get to know all of you! When I point to you, please state your **fullname** clearly so I can **input** your details into my **database**.

(Teacher collects and inputs student names)

Thank you! Now, let's quickly review our three **classroom expectations**:

1. Keep your eyes on the **instructor**.
2. Keep your hands off the **keyboard** and **mouse** when I am speaking.
3. Be **respectful** and work **peacefully**.

Are we ready to begin? Give me a big **thumbs-up!** Perfect, let's start!

PHASE 2: GOOGLE MEET SETUP & SCREEN SHARING (10 MINUTES)

Today, we are using **Google Meet** right here inside our physical lab. This allows you to view my **screen share** directly on your individual **monitors**.

Follow these steps **carefully**:

1. Open your **Chrome browser**.
2. Type **m-e-e-t-dot-g-o-o-g-l-e-dot-c-o-m** (**meet.google.com**) in the **address bar** and press **Enter**.
3. Click on **Join or start a meeting** and enter our **passcode: P-Y-T-H-O-N-1**.
4. Before clicking join, check your display name! You must follow our strict **naming format**: type your **class**, then a **dash**, followed by your **fullname**—for example, **7A - John Smith** or **8B - Sarah Jenkins**. This allows me to **identify** you and address you properly in the room.
5. Click **Join now!**

Look at your screens. You should now see my **presentation screen share!** When I demonstrate **algorithms** and code, you can follow along on your screen.

PHASE 3: TARGET CONCEPTS & GUIDED PRACTICE (40 MINUTES)

FOR GRADE 7 (Flowcharts & Intro to Python):

Please open your **textbooks** to **page ten** (10). We will examine pages 10 through 16.

Before writing actual **Python syntax**, we must learn how to design logic using **flowcharts**. A **flowchart** is a visual diagram of an **algorithm**.

Look at the **diagrams** on pages 10 to 16:

- **Ovals** represent the **Start** and **End** of a process.
- **Parallelograms** represent **Input** and **Output**.
- **Rectangles** represent **Processing** or actions.
- **Diamonds** represent **Decision points**—asking "Yes" or "No" questions.

Let's analyze the **worked examples** on page 14 together. Look at how the **arrows** direct the **execution flow**. When you write code in **Python**, your script follows this exact logical path step-by-step!

Now that you understand **flowcharts**, open a new **tab** and go to **c-o-d-e-x-dot-i-o** (**codex.io**). Log in with your **section code**, and start working on the **introductory Python games** to translate your flowchart logic into real code!

FOR GRADE 8 (Python Fundamentals: Hello World & Variables):

For Grade 8, we are jumping straight into text-based programming using **Python** on **codex.io**!

Open a new **tab** in your **browser**:

1. Go to **c-o-d-e-x-dot-i-o** (**codex.io**).
2. Click **Sign in** and enter our **section code: P-Y-8-C-O-D-E**.
3. Select your **fullname** and log in!

Your goal today is to complete two core chapters:

1. **Chapter 1: Hello World** — Learning how to use the **print()** **function** to display text output.
2. **Chapter 2: Variables** — Learning how to create **variables** to store **data types** like **strings**, **integers**, and **booleans**.

SELF-PACED GAMEPLAY (30 MINUTES - BOTH GRADES):

You may now begin your interactive challenges on **codex.io**.

Remember, programming is a **self-paced** journey. It is not a **race**. Focus on understanding the **logic** behind every line of code.

If you get **stuck**:

- **Option 1: Re-read** the instructions and error messages on your screen carefully.
- **Option 2:** Ask your **neighbor** sitting next to you **quietly**.
- **Option 3:** Raise your hand **patiently**, and I will check your code.

You have 30 minutes. Start **coding**!

(Students work independently while the teacher monitors progress via the online dashboard)

PHASE 4: LESSON SUMMARY & CLEANUP (5 MINUTES)

1. Summary & Closing (2.5 mins):

Time is up! Please **freeze** and place your hands on your desks!

Look at my **screen share** on **Google Meet** one last time. I am reviewing our **teacher dashboard**, and your progress today is **outstanding**!

- **Grade 7:** You successfully mapped complex logic using **flowcharts** and translated that thinking into **Python**!
- **Grade 8:** You mastered **syntax**, printed outputs with `print()`, and manipulated **variables** effectively!

Remember, learning to program is about developing **problem-solving** resilience. It does not matter which specific **stage** or level you reached today—what matters is that you **persevered** through errors and built real **computational thinking** skills. We will pick up right where you left off in our next **session**. Give yourselves a big **round of applause**!

2. Lab Cleanup (2.5 mins):

Now, let's leave our room clean and **organized**:

1. Click the red **phone icon** to leave the **Google Meet**.
2. Close your **Chrome browser** by clicking the red **X**.
3. Click **Start**, click **Power**, and click **Shut down** to power off your **computers**.
4. Place your **mouse**, **keyboard**, and **textbooks** neatly on your desk.
5. Push your **chairs** in **silently**.

Check your area for any **trash**. Line up at the door **quietly**. You were all **extraordinary** programmers today! See you next week! Goodbye, **class**!

Database	<i>/ˈdeɪ.tə.beɪs/</i>	Cơ sở dữ liệu	Stress on the first syllable (DAY).
Instructor	<i>/ɪnˈstrʌk.tə/</i>	Người hướng dẫn / Giáo viên	Stress on the second syllable (STRUK).

Respectful	/rɪ'spekt.fəl/	Tôn trọng	Soft /k/ sound in the middle; light /-fəl/ ending.
Peacefully	/'pi:s.fəl.i/	Trật tự / Yên bình	Long /i:/ sound; pronounced in 3 clear syllables.
Display name	/dɪ'spleɪ neɪm/	Tên hiển thị	Stress falls on PLAY .
Naming format	/'neɪ.mɪŋ ,fɔ:r.mæt/	Định dạng đặt tên	Clearly pronounce the final /t/ in "format".
Identify	/aɪ'den.tʃə.faɪ/	Nhận diện / Xác định	Stress on the second syllable (DEN).
Presentation	/.prez.ən'teɪ.ʃən/	Bài thuyết trình / Trình chiếu	Primary stress on the third syllable (TAY).
Flowchart	/'fləʊ.tʃɑ:rt/	Sơ đồ khối (Lớp 7)	Distinct /oʊ/ sound in "flow" and /tʃ/ in "chart".
Algorithm	/'æl.gə.rɪ.ðəm/	Thuật toán	Soft /ð/ sound; stress on the first syllable (AL).
Diagrams	/'daɪ.ə.græmz/	Sơ đồ / Biểu đồ	Pronounced in 3 syllables (DI-a-grams).
Parallelogram	/.pær.ə'leɪ.ə.græm/	Hình bình hành (Input/Output)	Stress on the third syllable (LEL).
Decision points	/dɪ'sɪʒ.ən pɔɪnts/	Điểm rẽ nhánh / Quyết định	Pronounce the /ʒ/ sound softly in "decision".

Execution flow	/ˌek.səˈkjuː.ʃən fləʊ/	Lưu thông thực thi lệnh	Stress on KYOO in "execution".
Syntax	/ˈsɪn.tæks/	Cú pháp ngữ pháp lập trình	Clear /ks/ ending sound.
Variables	/ˈver.i.ə.bəlz/	Các biến số (Lớp 8)	Stress on the first syllable (VER).
Function	/ˈfʌŋk.ʃən/	Hàm trong lập trình	Stress on the first syllable (FUNGK).
Data types	/ˈdeɪ.tə taɪps/	Các kiểu dữ liệu	Clear /ps/ sound at the end of "types".
Integers	/ˈɪn.tə.dʒəz/	Kiểu số nguyên	Soft /dʒ/ sound (like "j"); stress on IN .
Booleans	/ˈbuː.li.ənz/	Kiểu đúng/sai (True/False)	Long /uː/ sound; stress on BOO .
Interactive	/ˌɪn.təˈæk.tɪv/	Tương tác	Primary stress on the third syllable (AK).
Persevered	/ˌpɜː.səˈvɪrd/	Kiên trì / Nỗ lực	Stress on the third syllable (VEER).
Computational	/ˌkɑːm.pjəˈteɪ.ʃən.əl/	Tư duy máy tính / Thuật toán	Primary stress on TAY .
Extraordinary	/ɪkˈstrɔːr.dən.ər.i/	Phi thường / Tuyệt vời	5-syllable word; stress on STROR .

Grade 9 and Grade 10: Week 1

<https://github.com/codedex-io/html-101> and codex.io/python

While both grades share this week's technical curriculum, their academic contexts differ: **Grade 9** students are starting **Year 1** of their two-year IGCSE journey, whereas **Grade 10** students are in **Year 2** and preparing for their upcoming **May examinations**.

PHASE 1: CLASSROOM MANAGEMENT & ACADEMIC CONTEXT (5 MINUTES)

Good morning, **everyone**! Welcome to our **IGCSE ICT lab**!

Please stand next to your **desks**. Stretch your arms up high to the **ceiling**! Move them down to your **toes**! Now, sit down **quietly** and place your hands on your desks. Excellent!

Before we jump into our technical work today, I want to clarify our academic context:

- To my **Grade 9 students**: Welcome to **Year 1** of your **IGCSE ICT journey**! You have plenty of time to build your skills step-by-step over the next two years.
- To my **Grade 10 students**: You are now in **Year 2**! Your final **board examinations** are coming up next **May**, so every single practical session in this lab counts towards your final grade!

Let's quickly review our **classroom expectations**:

1. Keep your eyes on the **instructor**.
2. Keep your hands off the **keyboard** and **mouse** when I am speaking.
3. Be **respectful** and maintain a **peaceful** work environment.

Are we ready? Show me a big **thumbs-up**! Perfect, let's begin!

PHASE 2: GOOGLE MEET & IGCSE PAST PAPERS ANALYSIS (15 MINUTES)

Today, we will be using **Google Meet** right inside our lab so you can view my **screen share** directly on your **monitors**.

Please follow my instructions:

1. Open your **Chrome browser**.
2. Type **m-e-e-t-dot-g-o-o-g-l-e-dot-c-o-m** (**meet.google.com**) in the **address bar** and press **Enter**.
3. Click **Join or start a meeting** and enter our passcode: **I-G-C-S-E-1-0-1**.
4. Enter your name using our **naming format**: your **class**, a **dash**, and your **fullname**—for example, **9A - John Smith** or **10B - Sarah Jenkins**.
5. Click **Join now**!

Now, look at my **screen share** on **Google Meet**. I am distributing past **sample papers** for **IGCSE ICT**.

The **IGCSE ICT examination** consists of **three papers**:

- **Paper 1: Theory** — A written exam covering hardware, software, and systems analysis.
- **Paper 2: Document Production, Databases, and Presentations** — A practical exam testing document formatting and database management.
- **Paper 3: Data Analysis and Web Authoring** — A practical exam focusing on spreadsheets and **HTML/CSS web development**.

Look at these **sample papers**. How are you feeling about this structure? I know it looks challenging and formal right now, but do not worry! We will break everything down step-by-step.

PHASE 3: HTML DEMO, EVIDENCE DOCUMENTATION & CODEX.IO (35 MINUTES)

1. HTML Demonstration & Motivation (5 mins):

Look at my **screen share**. I am opening a few **web applications** that I personally built using **HTML**, **CSS**, and **JavaScript**.

Look at how these layout **elements**, text styles, images, and **interactive links** come together to form a full web application! It might look like magic now, but if you work hard and remain **diligent**, you will be able to build your own functional web applications in just a few weeks!

2. Evidence Document Workflow (10 mins):

In your **IGCSE Paper 2 and Paper 3 exams**, you are required to submit an **Evidence Document**. You will lose marks if you do not document your work properly!

Here is your process for today:

1. Go to **Google Classroom** and open the assigned **Google Docs** file titled **"Evidence Document - Week 1"**.
2. Every time you complete a task or write code, you must take a **screenshot** of your result and paste it directly into your **Google Docs** file.
3. If you do not know how to take a **screenshot** on your system, ask your **neighbor quietly** or search on the **internet**.
4. Once completed, **submit** your file on **Google Classroom**.

Important Note: Since this is our **first week**, I will be lenient if you are feeling a bit lost or **confused**. However, starting next lesson, mastering this **evidence submission** process is mandatory!

3. Codex.io Elements Module Practice (20 mins):

Now, open a new **tab** and go to **c-o-d-e-x-dot-i-o-slash-h-t-m-l** (codex.io/html). Today, your task is to complete the entire **Elements** module.

Let's briefly overview the core **HTML elements** you will master inside the game:

- **Structure & Block Elements:** <body> (document body), **headings** (<h1> to <h4>), and <p> (**paragraph**).
- **Text Formatting Elements:** (**bold**), <i> (**italic**), <u> (**underline**), <s> (**strikethrough**), and
 (**line break**).
- **Lists:** (**unordered list**), (**ordered list**), and (**list item**).
- **Media & Links:** (**image tag**) with the **src attribute**, and <a> (**anchor tag**) with the **href attribute**.
- **Special Syntax & Legacy Tags:** <marquee> (**scrolling text**—a **deprecated element**), and **HTML comments** (<!-- comment -->).

You have 20 minutes to work through **codex.io/html**, complete the **Elements** section, capture your **screenshots**, and submit your **Evidence Document** on **Google Classroom**. Start **coding**!

(Students work independently while teacher monitors progress on the dashboard)

PHASE 4: LESSON SUMMARY & LAB CLEANUP (5 MINUTES)

1. Summary & Closing (2.5 mins):

Time is up! Please **freeze** and place your hands on your desks!

Look at my **screen share** on **Google Meet** one last time. I can see your submissions coming in on **Google Classroom**, and I am thoroughly **impressed** with your focus!

- **Grade 9 students:** You have taken your very first steps into **IGCSE Web Authoring**! You now understand how raw **HTML tags** structure content on the web.
- **Grade 10 students:** You demonstrated great **discipline** today. Establishing this **evidence collection** routine now will make your final exam preparation smooth and stress-free!

Remember, web development is built line by line. It does not matter how far you got today—what matters is that you **persevered** and learned how to document your work like real **engineers**. We will pick up right here next session. Give yourselves a big **round of applause**!

2. Lab Cleanup (2.5 mins):

Now, let's leave our lab completely **organized**:

1. Click the red **phone icon** to leave the **Google Meet**.
2. Close your **Chrome browser** by clicking the red **X**.
3. Click **Start**, click **Power**, and click **Shut down** to power off your **computers**.
4. Place your **mouse**, **keyboard**, and **sample papers** neatly on your desk.

5. Push your **chairs** in **silently**.

Check your area for any **trash**. Line up at the door **quietly**. Excellent work today! See you next week! Goodbye, **class**!

TEACHER'S GLOSSARY

Examinations	/ɪgˌzæm.əˈneɪ.ʃənz/	Các kỳ thi	Primary stress on NAY .
Instructor	/ɪnˈstrʌk.tə/	Người hướng dẫn / Giáo viên	Stress on the second syllable (STRUK).
Respectful	/rɪˈspekt.fəl/	Tôn trọng	Soft /k/ sound in the middle; light /-fəl/ ending.
Peaceful	/ˈpiːs.fəl/	Trật tự / Yên bình	Long /i:/ sound; pronounced clearly.
Passcode	/ˈpæs.koʊd/	Mật khẩu truy cập	Clear /s/ in "pass" and /d/ at the end of "code".
Display name	/dɪˈspleɪ neɪm/	Tên hiển thị	Stress falls on PLAY .
Naming format	/ˈneɪ.mɪŋ ˈfɔːr.mæt/	Định dạng đặt tên	Clearly pronounce the final /t/ in "format".
Sample papers	/ˈsæm.pəl ˈpeɪ.pəz/	Đề thi mẫu	Clear /p/ sounds; stress on SAMP and PAY .
Theoretical	/ˌθiː.əˈreɪ.kəl/	Thuộc về lý thuyết (Paper 1)	Stress on RET ; starts with a soft /θ/ sound.
Web Authoring	/web ˈɑː.θə.ɪŋ/	Lập trình trang web (Paper 3)	Soft /θ/ sound in "authoring".
Diligent	/ˈdɪl.ə.dʒənt/	Chăm chỉ / Cần cù	Stress on the first syllable (DIL).

Evidence Document	/ˈev.ə.dəns ˈdɑː.kjə.mənt/	Tài liệu bằng chứng IGCSE	Stress on EV and DOC .
Screenshot	/ˈskriːn.ʃɑːt/	Ảnh chụp màn hình	Clear /k/ sound in "screen" and /t/ in "shot".
Mandatory	/ˈmæn.də.tɔːr.i/	Bắt buộc / Bắt tay vào nếp	Stress on the first syllable (MAN).
Paragraph	/ˈpær.ə.græf/	Đoạn văn (<p>)	Stress on the first syllable (PAR).
Unordered list	/ʌnˈɔːr.dəd list/	Danh sách không thứ tự ()	Clear /d/ at the end of "unordered".
Ordered list	/ˈɔːr.dəd list/	Danh sách có thứ tự ()	Stress on the first syllable (OR).
Attribute	/ˈæt.rə.bjuːt/	Thuộc tính thẻ (src, href)	Noun form: stress on the first syllable (AT).
Deprecated	/ˈdep.rə.keɪ.tɪd/	Đã cũ / Không khuyến khích	Stress on DEP ; 4 clear syllables.
Engineers	/ˌen.dʒəˈnɪrz/	Kỹ sư	Primary stress on the last syllable (NEERS).
Persevered	/ˌpɜː.səˈvɪrd/	Kiên trì / Nỗ lực	Stress on the third syllable (VEER).
Organized	/ˈɔːr.gə.naɪzd/	Ngăn nắp	Stress on the first syllable (OR).

Dưới đây là thống kê toàn bộ các thẻ (elements), thuộc tính (attributes) và kiến thức HTML có trong mã nguồn bạn đã cung cấp, được phân loại rõ ràng theo từng nhóm chức năng:

1. Các thẻ cấu trúc & Khối (Structure & Block Elements)

- `<body>`: Thẻ chứa toàn bộ nội dung hiển thị trên trang web.
- **Thẻ tiêu đề** (`<h1>`, `<h2>`, `<h3>`, `<h4>`): Dùng để tạo các cấp độ tiêu đề nội dung từ lớn đến nhỏ (mức độ ưu tiên/kích thước giảm dần từ `<h1>` đến `<h4>`).
- `<p>`: Thẻ đoạn văn (Paragraph) dùng để chứa khối văn bản.

2. Các thẻ định dạng văn bản (Text Formatting Elements)

- `<b>`: In đậm văn bản (Bold).
- `<i>`: In nghiêng văn bản (Italic).
- `<u>`: Gạch chân văn bản (Underline).
- `<s>`: Gạch ngang văn bản (Strikethrough / Deleted text).
- `<br>`: Ngắt dòng (Line break) trong đoạn văn.

3. Danh sách (Lists)

- `<ul>`: Danh sách không phân cấp / không thứ tự (Unordered List) - hiển thị dạng dấu đầu dòng (bullet points).
- `<ol>`: Danh sách có thứ tự (Ordered List) - hiển thị dạng đánh số 1, 2, 3....
- `<li>`: Mục con trong danh sách (List Item) - dùng bên trong cả `<ul>` và `<ol>`.

4. Thẻ đa phương tiện & Liên kết (Media & Links)

- `<img>`: Thẻ chèn hình ảnh (thẻ tự đóng).
 - **Thuộc tính** `src`: Đường dẫn URL dẫn đến tệp hình ảnh.
- `<a>`: Thẻ tạo thẻ liên kết (Anchor / Hyperlink).
 - **Thuộc tính** `href`: Địa chỉ web (đường dẫn URL) hoặc giao thức đặc biệt (ví dụ: `tel`: dùng để tạo liên kết gọi điện thoại nhanh).

5. Thẻ cổ điển / Thẻ động (Deprecated Elements)

- `<marquee>`: Thẻ tạo hiệu ứng chữ hoặc hình ảnh chạy cuộn qua lại trên màn hình (*Lưu ý: Thẻ này đã cũ và không còn được khuyến khích sử dụng trong tiêu chuẩn HTML5 hiện đại, nhưng xuất hiện trong ví dụ của bạn.*)

6. Cú pháp bỏ trợ

- **HTML Comments** (`<!-- ... -->`): Chú thích trong code HTML (trình duyệt sẽ bỏ qua không hiển thị nội dung này ra giao diện).

Bảng tổng hợp danh sách thẻ theo thứ tự bảng chữ cái

Thẻ HTML	Tên đầy đủ / Ý nghĩa
<!-- -->	Comment (Ghi chú)
<a>	Anchor (Thẻ liên kết)
	Bold (In đậm)
<body>	Document Body (Thân trang web)
 	Line Break (Xuống dòng)
<h1> – <h4>	Headings (Tiêu đề cấp 1 đến 4)
i	Italic (In nghiêng)
	Image (Hình ảnh)
	List Item (Mục danh sách)
<marquee>	Scrolling Text (Chữ chạy - thẻ cũ)
	Ordered List (Danh sách có thứ tự)
<p>	Paragraph (Đoạn văn)
<s>	Strikethrough (Gạch ngang chữ)
<u>	Underline (Gạch chân)
	Unordered List (Danh sách không thứ tự)

Grade 6: W1: PHASE 1: CLASSROOM MANAGEMENT & GROUP FORMATION 5mins

Good morning, **everyone!** Welcome to our **computer lab!**

Please stand next to your **desks**. Stretch your arms up high to the **ceiling!** Touch your **toes!** Now, sit down **quietly** and place your hands on your desks. Excellent!

Since there are **twenty students** in our class today, we are going to divide into **small groups** of **two to three students**.

Please turn to your **neighbors** and form your groups of two or three now.

(Students form groups)

Let's quickly review our three **classroom expectations**:

1. Keep your eyes on the **instructor**.
2. Keep your hands off the **keyboard** and **mouse** when I am speaking.
3. Be **respectful** and work **peacefully**.

Are we ready? Show me a big **thumbs-up!** Perfect, let's begin!

PHASE 2: LAB RULES POSTER PROJECT (20 MINUTES)

Look at my **presentation screen share**. Today, your main **task** is to create a colorful **classroom rules poster** on **A4 paper!**

Before you start drawing, your group must **register** your group name and selected **lab rules** on a piece of paper with me!

You can choose rules from our **sample lists**, or you can create your own **custom rules**:

1. Safety & Equipment Protection:

-  No food or drinks inside the lab!
-  Wash and dry your hands before touching equipment.
-  Do not touch or unplug electrical **cables**.
-  Handle the **mouse** and **keyboard** gently.
-  No running or **horseplay** in the lab!

2. Conduct & Behavior:

-  Wear headphones and keep your voices **peaceful**.
-  **Line up** quietly when entering or exiting the room.
-  No playing unauthorized games or browsing unknown websites!
-  Raise your hand when you need assistance from the **teacher**.

- 🤝 💻 **Collaborate** and share computers happily with your partners!

3. Cleanliness & End of Class:

- 🧹 🗑️ Clean up trash and keep your **workstation** tidy!
- 🪑 🚰 Push your **chairs** in neatly when you stand up.
- 💻 🔴 **Shut down** computers properly.
- 🎧 🖱️ Place keyboards, mice, and headphones back in their original **positions**.
- 🧳 🎒 Double-check all personal **belongings** before leaving!

Look at this **sample poster** on my screen. Notice how clear the **illustrations**, icons, and **typography** are!

Now, group leaders, come to the front, **register** your group name and chosen rules, collect your **A4 paper** and colored markers, and start **drawing**! You have 15 minutes. Begin!

(Students work on posters while teacher walks around providing feedback)

Three, two, one... time is up! Please hold up your **posters**! These look **extraordinary**! Give yourselves a **round of applause**!

PHASE 3: INTRO TO MAKECODE MICRO:BIT & FLASHING HEART (30 MINUTES)

Now, let's transition to physical computing and block programming!

Please open your **Chrome browser** on your computers:

1. Type **m-a-k-e-c-o-d-e-dot-m-i-c-r-o-b-i-t-dot-o-r-g** (makecode.microbit.org) into the **address bar** and press **Enter**.
2. Click on **New Project** and name it "**Flashing Heart**".

Before we start coding, look at my screen. Let's learn about the **hardware structure** of the **micro:bit**:

- **LED Grid:** A 5x5 **matrix** of 25 red LEDs to display images, text, and numbers.
- **Buttons:** Button A and Button B on the front for user **inputs**.
- **Processor & Sensors:** On the back, it contains a **microprocessor**, an **accelerometer** to detect motion, and a **compass**.
- **USB Port & Battery Connector:** To power and transfer code to the board.

Now, look at my screen share to watch this short **demonstration video** of the **Flashing Heart** project!

Notice how the heart icon appears, disappears, and repeats endlessly! This creates an **animation** effect using a **forever loop**!

Now, it's your turn to code it in **MakeCode**:

1. Drag a **show leds** block from the **Basic** category and place it inside the **forever** block. Draw a big heart!
2. Drag a **pause** block underneath it to hold the heart image.
3. Drag a second **show leds** block, but leave it blank (or draw a small heart).
4. Watch the virtual **micro:bit simulator** on the left side of your screen!

You have 15 minutes to complete and customize your **Flashing Heart animation**. Go, **programmers**!

(Students code while teacher assists)

Look at those glowing, beating hearts on your simulators! That was **phenomenal** work!

PHASE 4: LESSON SUMMARY & LAB CLEANUP (5 MINUTES)

1. Summary & Closing (2.5 mins):

Time is up! Please **freeze** and place your hands on your desks!

Today, we accomplished two major goals:

1. You established a strong foundation of lab **safety** and etiquette through your **custom posters**.
2. You explored the **micro:bit architecture** and successfully built your very first **Flashing Heart algorithm**!

Remember, learning computer science is a step-by-step **journey**. You showed great **creativity** and **collaborative** effort today. Give yourselves a big **round of applause**!

2. Lab Cleanup (2.5 mins):

Now, let's practice the very **lab rules** we drew today to leave our room completely **organized**:

1. Close your **Chrome browser** by clicking the red **X**.
2. Click **Start**, click **Power**, and click **Shut down** to turn off your **computers**.
3. Return all markers, pencils, and **A4 paper** to the front table.
4. Place your **mouse**, **keyboard**, and headphones back in their proper **positions**.
5. Push your **chairs** in **silently**.

Instructor	/ɪnˈstrʌk.tə-/	Người hướng dẫn / Giáo viên	Stress on the second syllable (STRUK).
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Respectful	/rɪˈspekt.fəl/	Tôn trọng	Soft /k/ sound in the middle; light /-fəl/ ending.
Peacefully	/ˈpiːs.fəl.i/	Trật tự / Yên bình	Long /i:/ sound; pronounced in 3 clear syllables.
Equipment	/ɪˈkwɪp.mənt/	Thiết bị	Stress on the second syllable (KWIP); clear final /t/.
Horseplay	/ˈhɔːrs.pleɪ/	Đùa giỡn, chạy nhảy gây nguy hiểm	Compound word; stress on HORS .
Conduct	/ˈkɔːn.dʌkt/	Nếp sống / Ý thức chấp hành	Noun form: stress on the first syllable (KON).
Collaborate	/kəˈlæb.ə.reɪt/	Hợp tác / Làm việc nhóm	Stress on the second syllable (LAB).
Workstation	/ˈwɜːk.steɪ.ʃən/	Góc / Màn làm việc	Compound word; clear /k/ between words.
Belongings	/bɪˈlɔːŋ.ɪŋz/	Tư trang / Đồ dùng cá nhân	Stress on the second syllable (LONG).
Illustrations	/ˌɪl.əˈstreɪ.ʃənz/	Hình ảnh minh họa	Primary stress on STRAY .

Typography	/taɪ'pɒ:grə.fi/	Nghệ thuật chữ / Định dạng chữ	Stress on the second syllable (PA).
Micro:bit	/'maɪ.krou.bi:t/	Bộ mạch vi điều khiển micro:bit	Stress on MY ; clear /t/ at the end.
Hardware	/'hɑ:rd.wer/	Phần cứng	Stress on HARD .
Architecture	/'ɑ:r.kə.tek.tʃə/	Cấu trúc / Kiến trúc phần cứng	Stress on the first syllable (AR); /k/ sound.
Matrix	/'meɪ.trɪks/	Ma trận (Lưới 5x5 LED)	Stress on MAY ; clear /ks/ ending.
Accelerometer	/ək'sel.ə'raɪ.mə.tə/	Cảm biến gia tốc / Cảm biến chuyển động	Primary stress on RA .
Microprocessor	/,maɪ.krou'prɑ:ses.ə/	Bộ vi xử lý trung tâm	Primary stress on PROS .
Simulator	/'sɪm.jə.leɪ.tə/	Trình giả lập	Stress on the first syllable (SIM).
Algorithm	/'æl.gə.rɪ.ðəm/	Thuật toán	Soft /ð/ sound; stress on AL .
Collaborative	/kə'læb.ə.ə.tɪv/	Có tính hợp tác	Stress on the second syllable (LAB).
Extraordinary	/ɪk'strɔ:r.dən.er.i/	Phi thường / Tuyệt vời	5-syllable word; stress on STROR .